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26 May 1992

SUPERSEDING
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MILITARY SPECIFICATION

POWER SUPPLY
25 MILLIMETER, MICROCHANNEL INVERTER

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the power supply for the Image Intensifier, Night Vision MX-9644/UV.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.e).

Beneficial comments (recommendations, additions, deletions) and pertinent data which may be of use in improving this document should be addressed to: HQ, U.S. Army Communications-Electronics Command, ATTN: AMSEL-ED-TM, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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SPECIFICATIONS

MILITARY

MIL-P-11268	-	Parts, Materials, and Processes Used in Electronic Equipment
MIL-B-49030/6	-	Battery, Dry, Type BA-3058/U
MIL-I-49040	-	Image Intensifier Assembly 25 Millimeter, Microchannel Inverter MX-9644/UV
MIL-B-49430/4	-	Battery, Primary, Lithium, Sulfur-Dioxide Type BA-5567/U

STANDARDS

MILITARY

MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirement for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristic Measurement of
MIL-STD-781	-	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-810	-	Environmental Test Methods
MIL-STD-883	-	Test Methods and Procedures for Microelectronics

HANDBOOKS

MILITARY

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment
MIL-HDBK-781	-	Reliability Test Methods, Plans, and Environments for Engineering Development Qualification, and Production

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5095.)

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.e).

DRAWINGS

USACECOM

SM-D-850290	-	Power Supply, 25 Millimeter, Microchannel Inverter
SM-D-850299	-	Housing, Image Intensifier
SM-D-771336-1	-	Electronics Control Assembly

ARMCOM

11747171	-	Regulator Assembly

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.e).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

S1.2-1962	-	American National Standard Method for the Physical Measurement of Sound

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Construction. Unless otherwise specified, all parts, materials and processes used in construction of the power supply shall be in accordance with SM-D-850290.

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3.2 Weight. The power supply shall not exceed 63 grams in its total weight.

3.3 Qualification. Unless otherwise specified by the contract or purchase order, Power Supplies furnished under this specification shall be products approved as qualified products (see 3.12.16 and 6.3).

3.3.1 Initial Production Testing. When specified in the contract or purchase order, the contractor shall furnish the IPT units in accordance with 4.3 (see 6.2.b).

3.4 Materials. The selection of the class, grade, or type shall be such that the part material will perform its intended function when assembled in the power supply. Parts and materials not specified shall be selected by the contractor and shall be subjected to all the provisions of MIL-P-11268 and as specified herein. Hybrid microelectronics or integrated electronics shall be tested to the class B level of Method 5008/5004 of MIL-STD-883.

3.5 Components. The components of the power supply are identified as the regulated oscillator module and high voltage multiplier module. The oscillator/multiplier interface connection shall be in accordance with the applicable drawing. The maximum allowed dc voltage at the interface connection shall be 1,200 Vdc. Maximum alternating current (ac) voltage at the interface connection shall be 550 volts peak to peak.

3.6 Operation characteristics. Unless otherwise specified herein, the power supply shall have the following performance characteristics at ambient conditions (see 3.12.1). The power supply shall meet all operation characteristics when powered by battery BA-3058/U conforming to MIL-B-49030/6, BA-5567/U, conforming to MIL-B-49430/4 or an equivalent power source (see 3.12.15). Connection between the power supply and the intensifier module of Image Intensifier, Night Vision MX-9644/UV are given in Figure 1. Polarity of each output shall be in accordance with Figure 2.

3.6.1 Voltage adjustment. The power supply shall operate in conjunction with an ON/OFF external gain adjust control (EGAC) circuit as shown in Figure 1. The maximum gain position of the EGAC shall be obtained with the switch (S1) closed and the wiper rotated CW to position 3. The minimum gain position of the EGAC shall be obtained with the switch (S1) closed and the wiper rotated CCW to position 1.

3.6.1.1 Maximum gain.

a. The microchannel plate input voltage (V_{10}) shall be not greater than 1,200 Vdc when the adjustments R_1 and R_2 are at their maximum gain settings with the following conditions applied to the power supply.

(1) Input voltage is 3.0 Vdc

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(2) The load at the output 10 is 350 megohm

(3) All other outputs shall be at no load (see 3.12.6)

(4) The EGAC is at its maximum gain position.

b. V_{10} shall be less than 700 Vdc when the adjustment R_1 varies from its maximum to minimum gain setting at the conditions as specified in the step (a) above. The adjustment R_2 shall remain at its maximum gain setting during the variation of R_1.

c. V_{10} shall be greater than 1,000 Vdc when the adjustment R_1 varies from its minimum to maximum gain setting at the following conditions applied to the power supply. The adjustment R_2 shall remain at its maximum gain setting during the variation of R_1.

(1) Input voltage is 2.0 Vdc

(2) The load at the output 10 is 75 megohm

(3) All other outputs shall be at no load.

(4) The EGAC is at its maximum gain position.

3.6.1.2 Minimum gain.

a. With the EGAC at its minimum gain position and other conditions remaining as specified in 3.6.1.1.b, V_{10} shall be greater than or equal to 90 percent of its preset value per 3.6.1.1.b. When the adjustment R_2 varies from its maximum to minimum gain setting, V_{10} shall be less than 78 percent of its preset value per 3.6.1.1.b.

b. With the EGAC at its minimum gain position and other conditions remaining as specified in 3.6.1.1.c, V_{10} shall be greater than or equal to 90 percent of its preset value per 3.6.1.1.c. When the adjustment R_2 varies from its maximum to minimum gain setting, V_{10} shall be less than 78 percent of its preset value per 3.6.1.1.c.

3.6.2 Automatic brightness control (ABC) current characteristics.

3.6.2.1 ABC start current. When the power supply is operated with the EGAC at maximum gain position, the ABC start current (see 3.12.3) shall be adjustable within the range as follows:

a. The ABC start current at the output 13 shall be adjustable from 200 to 450 nanoamperes (nA) by the adjustment R_3.

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b. The ABC start current at the output 12 shall be adjustable from 200 to 700nA by the adjustment R_3.

c. The ABC start current in the step (a) and (b) above shall not vary more than ± 10 percent when the input voltage is changed from 3.0 to 2.0 Vdc.

3.6.2.2 ABC finish current. The ABC finish current (see 3.12.2) with the EGAC at maximum and minimum gain positions shall be not greater than 50.0 nA above the ABC start current of 3.6.2.1.

3.6.3 Output regulation.

3.6.3.1 Normal mode.

a. With the EGAC at maximum gain position, and as the load current varies within the range of 0 to 40 percent of the ABC start current, V_{10} shall not vary more than ± 20.0 Vdc from its initial preset value of Table I. (The initial preset value is defined as the output voltage of V_{10} when the load current is equal to 25nA or less and the applied input voltage is 3.0 Vdc).

b. With EGAC at mimimum gain position, and as the load current varies within the range of 0 to 40 percent of the ABC start current, V_{10} shall not vary more than ± 25.0 Vdc from its established voltage difference specified in 3.6.1.2.

c. The output voltage requirements at outputs 5, 6, 7, 8, 9, 12 and 13 (see Figure 2) shall be within the prescribed limits of Table I. These requirements shall be met when the power supply is operated with the EGAC at maximum and at minimum gain positions.

3.6.3.2 High light resolution mode. The output voltages V_5 V_6, V_7, V_8, V_9, V_{10}, V_{12}, and V_{13} shall be as specified in Table II. These requirements shall be met when the power supply is operated with the EGAC at maximum and minimum gain positions.

3.6.3.3 Bright source protection mode.

a. With the EGAC at maximum position, V_{10} shall not exceed its preset value of 3.6.3.1.a by more than 50.0 Vdc.

b. The output voltages V_5 V_6, V_{12} and V_{13} shall be as specified in Table III. These requirements shall be met when the power supply is operated with the EGAC at maximum and minimum gain positions.

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Table I. Output regulation, normal mode.

Output Terminal	Temperature		
	+52°C	+23°C	-54°C
5	2,250 ≤ V5 ≤ 3,500Vdc at no load	2,500 ≤ V5 ≤ 3,500Vdc at no load	2,250 ≤ V5 ≤ 3,500Vdc at no load
6	1,500 ≤ V6 ≤ 2,000Vdc at no load	1,600 ≤ V6 ≤ 1,900Vdc at no load	1,500 ≤ V6 ≤ 2,000Vdc at no load
7	0.51 V6 ≤ V7 ≤ 0.6 V6 at no load	0.54 V6 ≤ V7 ≤ 0.58 V6 at no load	0.51 V6 ≤ V7 ≤ 0.6 V6 at no load
8	0.8 V5 ≤ V8 ≤ 0.9 V5 at no load	0.84 V5 ≤ V8 ≤ 0.9 V5 at no load	0.8 V5 ≤ V8 ≤ 0.9 V5 at no load
9	0.58 V6 ≤ V9 ≤ 0.67 V6 at no load	0.6 V6 ≤ V9 ≤ 0.65 V6 at no load	0.58 V6 ≤ V9 load
10	V10 at +52°C shall be within the range of +25Vdc and -20Vdc of the initial preset value set at +23°C. The load shall be 75% ± 2% of the load at +23°C.	700 ≤ V10 ≤ 1,000Vdc for the loads with the range of 75 to 350 megohms.	V10 at -54°C shall be within the range of ± 20Vdc of the initial preset value set at +23°C. The load shall be 215% ± 2% of the load at +23°C.
12	4,650 ≤ V12 ≤ 5,500Vdc, and V12 at +54°C shall be within ± 250Vdc of the preset value at +23°C for a load current less than 40% of ABC start current.	4,800 ≤ V12 ≤ 5,300Vdc for a load current less than 40% of the 2ABC start current.	4,650 ≤ V12 ≤ 5,500Vdc and V12 at -54°C shall be within ± 250 Vdc of the preset value at +23°C for a load current less than 40% of ABC start current.
13	5,500 ≤ V13 ≤ 6,250Vdc, and V13 at +54°C shall be within ± 250Vdc of the preset value at +23°C for a load current less than 40% of ABC start current.	5,750 ≤ V13 ≤ 6,250Vdc for a load current less than 40% and V13 of the ABC start current	5,500 ≤ V13 ≤ 6,250Vdc and V13 at -54°C shall be within ± 250Vdc of the preset value at +23°C for a load current less than 40% of ABC start current.

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TABLE II. Output regulation, high light resolution mode.

Output Terminal	Temperature +52°C	+23°C	-54°C
5	V5 ≥ 500Vdc for a load current greater than 2.5 microamperes (uA) and less than 2.6 uA.		
6	V6 ≥ 400Vdc for a load current greater than 2.5 uA and less than 2.6 uA.		
7	V7 ≥ 300Vdc at no load		
8	0.4 V5 ≤ V8 ≤ 0.8 V5 at no load		
9	0.4 V6 ≤ V9 ≤ V6 at no load		
10	V10 ≤ 200Vdc. The load be 75%, ± 2% of the load at +23°C.	V10 ≤ 200Vdc for the load between 75 to 350 megohms.	V10 ≤ 200Vdc. The load shall shall be 215%, ± 2% of the load at +23°C.
12	For a load current greater than ABC finish current, V13 shall be ± 250 Vdc of the preset value at +23°C and also meet the requirement of 4,650 ≤ V12 ≤ 5,500Vdc.	4,800 ≤ V12 ≤ 5,300Vdc for a load current greater than the ABC finish current.	For a load current greater than ABC finish current, V12 shall be ± 250Vdc of the preset value at +23°C and also meet the requirement of 4,650 ≤ V12 ≤ 5,500Vdc.
13	For a load current greater than ABC finish current, V13 shall be ± 6250Vdc of the preset value at +23°C and also meet the requirement such that 5,500 ≥ V13 ≤ 6,250Vdc	5,750 ≤ V13 ≤ 6,250Vdc for a load current greater than ABC finish current	For a load current greater than ABC finish current, V13 shall be ± 250Vdc of the preset value at +23°C and also meet the requirement such that 5,500 ≤ V13 ≤ 6,250VDC

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TABLE III. Output regulation, bright source protection mode.

Output Terminal	Temperature +52°C	+23°C	-54°C
5	0 ≤ V5 ≤ 6.0Vdc for a load current greater than 2.6 uA and less than 6 uA		
6	0 ≤ V6 ≤ 8.0Vdc for a load current greater than 2.6 uA and less than 8 uA		
12	V12 ≤ 5,500Vdc at no load	V12 ≤ 5,300Vdc at no load	V12 ≤ 5,500Vdc at no load
13	V13 ≤ 6,250Vdc at no load	V13 ≤ 6,250Vdc at no load	V13 ≤ 6,250Vdc at no load

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3.6.4 Input current characteristics.

3.6.4.1 Maximum peak starting (MPS) and maximum peak steady state (MPSS). The MPS input current shall not exceed 300 milliampere (mA) and the MPSS input current shall not exceed 100 mA.

3.6.4.2 Maximum average steady state (MASS). The MASS input current shall not exceed 32mA.

3.6.5 Rise time. The rise time (see 3.12.7) of the power supply shall be not greater than 3 seconds, when the power supply is operated with the intensifier module from the Image Intensifier, Night Vision MX-9644/UV. The voltage overshoot of V_{10} shall not exceed 20 Vdc of its normal mode steady state value.

3.6.6 Response time (time dependent bright source). When the power supply is operating the intensifier module from Image Intensifier, Night Vision MX-9644/UV, which is subjected to a periodic pulsed input illumination in such a manner as to produce photocurrents not less than those specified in profile (a) and (b) as shown in Figures 3 and 4, the power supply shall have the following response time with the EGAC at the maximum position.

3.6.6.1 Recovery time, profile (a). The power supply shall recover to within 10 volts of its steady-state output voltage in 200 milliseconds or less after the end of the applied illumination, profile (a). Voltage overshoot of V_{10} shall be not greater than 20 Vdc of its normal mode steady state value.

3.6.6.2 Recovery time, profile (b). The power supply shall recover to within 10 volts of its steady-state output voltages in 75 milliseconds or less after the end of the applied illumination in profile (b). Voltage overshoot of V_{10} shall be not greater than 20 Vdc of its normal mode steady state value.

3.6.7 Ripple voltage.

a. The peak to peak ripple voltage at each output shall not exceed 3 percent (5 percent for V_{10}) of its steady state value when the power supply is operated in the normal mode.

b. The peak to peak ripple voltage shall not exceed 6 volts for V_5 and V_6 when the power supply is operated in the bright source protection mode.

3.6.8 Audio noise. Maximum sound pressure levels emanating from the power supply measured at a distance of 6 inches, shall not exceed the values tabulated below, against a background noise level at least 10 decibels below the sound pressure level to be measured. This requirement shall not be applicable for power supplies whose operating frequencies are greater than 22.4kHz.

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Frequency		
Center (Hz)	Octave Band (Hz)	Maximum Sound Pressure Level (dB: ref 0.0002 microbars)
8000	5600-11200	17
16000	11200-22400	16

3.6.9 Operational stability.

a. The power supply shall not exhibit arcing. There shall be no evidence of fluctuation or oscillation of the MASS input current greater than ± 0.5 mA.

b. V_{10} shall not fluctuate greater than ± 5 volts from its steady state value and shall not drift greater than ± 5 volts from its steady state value. Periodic fluctuations not greater than the ripple voltage requirements (see 3.6.7) shall be allowed only at the V_{10} oscillator frequency. All other fluctuations shall be random.

3.6.10 Operational safety.

3.6.10.1 Short circuit protection. While operating, the power supply shall not be damaged (see 3.12.4), when subjected to a direct short circuit of the high voltage output terminals to ground for at least one minute.

3.6.10.2 Reverse polarity. The power supply shall withstand a reverse polarity input voltage of 3.0 Vdc for at least 10 minutes. When the correct polarity of 3.0 Vdc input is applied to the power supply, the power supply shall conform to the requirement of 3.6.4.2.

3.6.11 Electromagnetic interference. The power supply shall comply with the following Class A3 requirements of MIL-STD-461: RE01, RE02, RS01, RS02 and RS03 except that the distance for measuring RE01 shall be 42 centimeter (cm). A change in MASS input current greater than 1.0 mA during the susceptibility tests RS01, RS02 and RS03 shall constitute the degradation of power supply performance.

3.7 Environmental conditions.

3.7.1 Temperature operation. The power supply shall not be damaged, (see 3.12.4) shall operate throughout the range of -54°C to +52°C, and shall meet the following performance requirements at the temperature specified. The power supply shall also meet the operation characteristics when powered by a battery BA-3058/U or BA-5567/U, and by an equivalent power source (see 3.12.15) as shown in Figure 1. Unless otherwise specified, these requirements shall be met with EGAC at maximum and minimum gain positions.

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3.7.1.1 Temperature operation at +52°C.

a. Output regulation:

(1) Normal mode - With the EGAC at maximum gain position, V_{10} shall be in accordance with Table I. With the EGAC at minimum gain position, V_{10} shall be in accordance with 3.6.3.1.b. The output voltage requirements of V_5 V_6, V_7, V_8, V_9, V_{12}, and V_{13} shall be in accordance with Table I with the EGAC at maximum and also at minimum gain position.

(2) High light resolution mode - The output regulation requirements shall be in accordance with Table II.

(3) Bright source protection mode - The output regulation requirements shall be in accordance with Table III.

b. For any discrete position of the adjustment R_3 meeting the requirement of 3.6.2.1, the ABC start current at +52°C shall not vary more than ± 30 percent of the corresponding value at ambient condition with 3.0 Vdc input. The ABC finish current shall be not greater than 50.0nA above the ABC start current at +52°C.

c. The rise time shall be not greater than 4 seconds. The voltage overshoot of V_{10} shall not exceed 20 Vdc of its normal mode steady state value (see 3.6.3.1).

d. The MASS input current shall be not greater than 35mA.

e. Operational stability requirements of 3.6.9 shall be met.

3.7.1.2 Temperature operation at -17.8°C.

a. The rise time shall be not greater than 3 seconds.

b. Operational stability requirement of 3.6.9 shall be met.

3.7.1.3 Temperature operation at -54°C.

a. Output regulation.

(1) Normal mode - With the EGAC at maximum gain position, V_{10} shall be in accordance with Table I. With the EGAC at minimum gain position, V_{10} shall be in accordance with 3.6.3.1.b. The output voltage requirements of V_5 V_6, V_7, V_8, V_9, V_{12}, and V_{13} shall be in accordance with Table I with the EGAC at maximum and also at minimum gain position.

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(2) High light resolution mode - The output regulation requirements shall be in accordance with Table II.

(3) Bright source protection mode - The output regulation requirements shall be in accordance with Table III.

b. For any discrete position of the adjustment R_3 meeting the requirement of 3.6.2.1, the ABC start current at -54°C shall not vary more than ± 30 percent of the corresponding value at ambient condition with 3.0 Vdc input. The ABC finish current shall be not greater than 50.0nA above the ABC start current at -54°C.

c. Rise time shall be not greater than 4 seconds. The voltage overshoot of V_{10} shall not exceed 20 Vdc of its normal mode steady state value (see 3.6.3.1).

d. The MASS input current shall be not greater than 25mA.

e. Operational stability requirements of 3.6.9 shall be met.

3.7.2 Environmental temperature stress. The power supply shall neither be damaged nor exhibit degradation of performance (see 3.12.4 and 3.12.17) after the storage or thermal shock environments specified herein.

3.7.2.1 High temperature stress. Storage at +68°C.

3.7.2.2 Low temperature stress. Storage at -62°C.

3.7.2.3 Temperature shock. Temperature changes occuring within 5 minutes in the temperature range of +68°C to -62°C.

3.7.3 Vibration. The power supply shall not be damaged after being rigidly mounted, and subjected to a 2.5 g's (see 3.12.5) peak sine wave vibration over a frequency range of 10 Hz through 3500 Hz along the longitudinal and transverse axes of the power supply.

3.7.4 Mechanical shock.

3.7.4.1 Mechanical shock I. The power supply shall not be damaged (see 3.12.4) when subjected to 6 shock impacts applied parallel to the radial axis and 6 shock impacts applied perpendicular to the radial axis of the assembly. The shock impacts shall be nominal half sine wave pulse having a minimum of 500 g's peak value and a duration of 0.3, ± 0.05 milliseconds measured between the 10 percent values of the peak amplitude.

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3.7.4.2 Mechanical shock II. The power supply shall not be damaged (see 3.12.4) when subjected to 6 mechanical shock pulses along the perpendicular axis and 6 mechanical shock pulses along the parallel axis at an amplitude of 140 g's, ± 10 g's half-sine wave peak for 9 milliseconds, ± 10 percent duration.

3.7.5 Humidity. The power supply shall not be damaged (see 3.12.4) when exposed to 95 percent relative humidity at 65°C.

3.8 Burn-in/environmental stress screening (ESS). Each power supply shall be set to standard operating setting (see 3.12.9) and shall undergo burn-in. Unless otherwise specified, no adjustment shall be allowed during or after the burn-in.

3.9 Reliability. The power supply shall have a lower test (θ_1), mean time between failure (MTBF) of 9000 hours.

3.10 Identification marking. Each power supply shall be marked for identification in accordance with drawing SM-D-850290. The marking shall include a coded acceptance date. The first 2 numbers of the code shall be the last 2 digits of the year. The last 2 numbers shall be 2 digits indicating the calendar week of the year (01 through 52). Reading from left to right (or top to bottom), the code number shall designate the year and week of acceptance in that order.

3.11 Workmanship. The power supply shall conform to MIL-STD-454. Requirement 9. The power supply shall be free from burrs, sharp edges, potting voids, defects, scratches which may affect power supply performance. It shall also be free of grease, oil, dust, solder flux or any other conductive film.

3.12 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.12.1 Ambient conditions. Ambient conditions are defined as follows:

Temperature	23°C, ± 10°C (73°F, ± 18°F)
Relative humidity	50 percent, ± 30 percent
Atmosphere pressure	725, +50mm of mercury/ -75mm of mercury (28.5, +2.0 inches of mercury/ -3.0 inches of mercury)

3.12.2 ABC finish current. The load current at output 12 (or 13) that results in a reduction of the absolute value of V_{10} to not greater than 200 Vdc.

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3.12.3 ABC start current. The load current at output 12 (or 13) that results in a change of V_{10} from its initial value at no load condition to 90 percent of that value.

3.12.4 Damage. Damage is defined as:

a. Breakage, loosening, shift, deformation, failure of finish, hardware, connection or component.
b. Increase or fluctuation in input current, corona, arcing, voltage instability or electrical failure or malfunctioning.

c. Any deterioration in quality or operation ability per the requirements of this specification.

3.12.5 "g". "g" is a unitless measure of the quantity of units of force associated with the earths gravity that a object is subjected to.

3.12.6 No load. The output current whose values do not exceed 25 nanoamperes.

3.12.7 Rise time. The time required for V_{10} to settle within 10 volts of its steady state value after the input voltage has been applied.

3.12.8 Steady state. The condition in which each output voltage remains constant after initial transient or fluctuation conditions have disappeared.

3.12.9 Standard operating setting. The standard operating setting shall be obtained as follows:

a. All parameters are shown in Figure 1.

b. Input voltage shall be 3.0, ± 0.05 Vdc.

c. Unless otherwise specified, load between outputs 10 and 11 shall be 75 megohm, and all other outputs shall be at no load condition.

d. With the EGAC at maximum gain position, V_{10} shall be equal to 825, ± 25 Vdc.

e. With the EGAC at minimum gain position the absolute value of V_{10} shall be equal to 150, ± 25 Vdc less than the absolute value of V_{10} with the EGAC at maximum gain position.

f. ABC start current of V_{13} shall be equal to 350, ± 50nA.

3.12.10 Low voltage operating setting. The low voltage operating setting shall be obtained as follows:

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a. All parameters are shown in Figure 1.

b. Input voltage shall be 3.0, ± 0.05 Vdc.

c. Unless otherwise specified, load between outputs 10 and 11 shall be 75 megohm and all other outputs shall be at no load conditions.

d. With the EGAC at maximum gain position, V_{10} shall be equal to 675, ± 25 Vdc.

e. With the EGAC at minimum gain position the absolute value of V_{10} shall be equal to 100, ± 25 Vdc less than the absolute value of V_{10} obtained with the EGAC at maximum gain position.

f. ABC start current of V_{13} shall be equal to 225, ± 25nA.

3.12.11 High voltage operating setting. The high voltage operating setting shall be obtained as follows:

a. All parameters are shown in Figure 1.

b. Input voltage shall be 3.0, ± 0.05 Vdc.

c. Unless otherwise specified, load between outputs 10 and 11 shall be 75 megohm and all other outputs shall be at no load condition.

d. With the EGAC at maximum gain position, V_{10} shall be within the range of 975 to 1,000 Vdc.

e. With the EGAC at minimum gain position, the absolute value of V_{10} shall be within the range of 150 to 175 Vdc less than the absolute value of V_{10} with the EGAC at maximum gain position.

f. ABC start current of V_{12} shall be equal to 675, ± 25nA.

3.12.12 Room temperature. Room temperature is defined as 23°C, ± 3°C.

3.12.13 Photocathode current. A signal induced current at output 5 (or 6), when the cathode of MX-9644/UV is illuminated.

3.12.14 Pattern failure. The occurrence of two (2) or more failures of the same part in identical or equivalent application whose combined failure rate exceeds that predicted. When two (2) or more dependent failures of the same part occur whose combined failure rate exceeds that predicted and which are due to more than one independent failure, the dependent failures are relevant pattern failures. Unless a Government approved Reliability Mathematic Model is available, the failure rates found in MIL-HDBK-217 shall be used.

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3.12.15 Equivalent power source. An equivalent power source is defined as the power source which will provide 2.0 to 3.0 Vdc to the power supply. The equivalent power source shall meet the output impedance and frequency ranges of electronic control assembly (drawing SM-D-771336-1, NSN: 5855-01-027-1564) of AN/VVS-2 and regulator assembly (ARMCOM drawing 11747171) of M32-E1 Periscope.

3.12.16 Qualified product. A product that has successfully met all the requirements of an IPT or First Article witnessed by the CECOM, Concurrent Engineering Directorate or its designated representative. Should the supplier make substantial changes in the process, personnel, location, or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, an IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characteristics of the product affected by the changes.

3.12.17 Degradation of performance. Any deterioration in quality , or failure to operate in accordance with the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Initial Production Testing (4.3).
b. Quality conformance inspection (4.5).

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4.3 Initial Production Testing Samples. Unless otherwise specified in the contract or purchase order the contractor shall furnish 15 power supplies from the first month's production for IPT inspection (see 6.2.b).

4.3.1 Inspection. Each IPT power supply shall be examined in accordance with Table IV. Presence of one or more defects shall be cause of rejection of that power supply.

TABLE IV. Inspection.

Defects	Requirement Paragraph
Components/materials missing or not as specified	3.4, 3.5
Construction not as specified	3.1
Identification not as specified	3.10
Workmanship not as specified	3.11

4.3.2 Initial Production Testing.

4.3.2.1 Initial Production Testing, Subgroup 1. Following successful completion of the inspections specified in 4.3.1, each IPT power supply shall be subjected to all tests of Table V, Subgroup 1. The tests of Table V, Subgroup 1 shall be performed in the order listed. Unless otherwise specified, the tests of Table V, Subgroup 1 shall be performed with the power supply at the standard operating setting (see 3.12.9). Failure of any power supply to pass all tests in Table V, Subgroup 1 shall be cause for rejection of that power supply and may be cause for termination of IPT testing.

4.3.2.2 Initial Production Testing, Subgroup 2. Five power supplies which have successfully passed the tests specified in 4.3.2.1 shall be subjected to the tests of Table V, Subgroup 2, at the high voltage operating setting (see 3.12.11). An additional 5 power supplies which have successfully passed the tests specified in 4.3.2.1 shall be subjected to the tests of Table V, Subgroup 2, while at the low voltage operating setting (3.12.10). Failure of any power supply to pass all tests in Table V, Subgroup 2 shall be cause for rejection of that power supply and may be cause for termination of IPT testing.

4.3.2.3 Initial Production Testing, Subgroup 3. Five power supplies which have successfully passed the tests specified in 4.3.2.1 shall be subjected to the tests of Table V, Subgroup 3 while at the standard operating setting. Two of the five shall be subjected to EMI tests. Failure of any power supply to pass all tests in Table V, Subgroup 3 shall be cause for rejection of that power supply and may be cause for termination of IPT testing.

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4.3.2.4 Initial Production Testing, Subgroup 4. Ten power supplies which have successfully passed the tests specified in 4.3.2.1 and 4.3.2.2 shall be subjected to the reliability test of Table V, Subgroup 4. Failure to meet the accept/reject criteria of the reliability test shall be cause for failure of the reliability test and may be cause for termination of IPT testing.

4.3.2.5 Disposition of IPT samples. IPT samples shall be shipped to the CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703 upon completion of all IPT testing.

TABLE V. Initial Production Testing.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Burn-in/ESS	3.8	4.8
ABC current characteristics*	3.6.2	4.6.2
Output regulation*	3.6.3	4.6.3
Input current characteristics*	3.6.4	4.6.4
Rise time*	3.6.5	4.6.5
Response time*	3.6.6	4.6.6
Operational stability*	3.6.9	4.6.9
Operational safety*	3.6.10	4.6.10
Temperature operation	3.7.1	4.7.1
Voltage adjustments*	3.6.1	4.6.1
Weight	3.2	4.6.12
Subgroup 2		
Environmental temperature stress	3.7.2	4.7.2
Vibration*	3.7.3	4.7.3
Mechanical shock - I*	3.7.4.1	4.7.4.1
Mechanical shock - II*	3.7.4.2	4.7.4.2
Humidity	3.7.5	4.7.5
Subgroup 3		
Ripple voltage*	3.6.7	4.6.7
Audio noise*	3.6.8	4.6.8
Electromagnetic interference*	3.6.11	4.6.11
Subgroup 4		
Reliability	3.9	4.9
* Test performed at +23°C		

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4.4 Inspection procedures. Tests shall be conducted with the test procedures specified herein. Unless otherwise specified the following conditions shall apply throughout the specification:

a. Tests shall be performed at ambient conditions (see 3.12.1).

b. The applied input voltage shall be 3.0 volts dc.

c. Tolerance on applied input voltages shall be ± 0.05 volts dc.

d. Test chambers used for environmental temperature tests shall maintain the temperature within ± 2°C.

e. All meters used during tests shall have been calibrated within 30 days prior to start of testing, and thereafter as required, and at intervals not to exceed 6 months. All meters shall be accurate within 1.0 percent of full scale reading, except the meter used to measure V_{10} which shall be accurate within 0.5 percent of full scale reading.

f. Tests shall be performed with the power supply mounted on an approved simulated tube body operated at ground potential and in an approved simulated grounded test housing. Measurements shall be performed utilizing the power supply test circuit (Figure 2), or its equivalent. Unless otherwise specified the following details shall apply.

(1) All switches shall be in the open position except switch S1.

(2) The EGAC shall be a maximum gain position.

g. Tests shall be performed at the standard operating setting (see 3.12.9).

4.5 Quality conformance inspection.

4.5.1 Group A inspection. Each power supply shall be inspected for the attributes as specified in Table IV and VI. The tests of Table VI, shall be performed sequentially as listed. Failure of a power supply to pass all inspections of Table VI shall be cause to reject the power supply. All tests shall be performed at +23°C except burn-in.

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TABLE VI. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Burn-in/ESS	3.8	4.8
ABC current characteristics	3.6.2	4.6.2
Output regulation	3.6.3	4.6.3
MASS input current	3.6.4.2	4.6.4.2

4.5.2. Group B inspection. Power supplies which have passsed Group A inspections in accordance with 4.5.1 shall be formed into lots by the contractor and shall be subjected to Group B inspections of Table VII. Sampling shall be as specified in the contract or purchase order (see 6.2.d). Tests for temperature operation at +52°C and at -54°C shall be performed prior to the test for voltage adjustment. At the completion of the Group B inspection, all power supplies shall be at the standard operation setting
(see 3.12.9).

TABLE VII. Group B inspection.

	Requirement paragraph	Test paragraph
Temperature operation at +52°C	3.7.1.1	4.7.1.1
Temperature operation at -54°C	3.7.1.3	4.7.1.3
Voltage adjustment	3.6.1	4.6.1
Dimensions	SM-D-850290	SM-D-850290
(a) Width		
(b) Inside radius		
(c) Outside radius		
(d) Location of terminals		

4.5.3 Group C inspection. This inspection shall consist of the tests specified in Table VIII and shall be performed on power supplies from lots which have passed Group B inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling of Group C inspections. Sampling plans for Group C tests shall be as specified in the contract or purchase order (see 6.2.d).

4.5.3.2 Group C failures. Actions required in the event of a Group C failure shall be as specified in the contract or purchase order (see 6.2.c).

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TABLE VIII. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1 - High Voltage		
Operational safety	3.6.10	4.6.10
Environmental temperature stress	3.7.2	4.7.2
Vibration	3.7.3	4.7.3
Mechanical shock I	3.7.4	4.7.4
Mechanical shock II	3.7.4.1	4.7.4.1
Humidity	3.7.5	4.7.5
Subgroup 2 - Low Voltage		
Operational safety	3.6.10	4.6.10
Environmental temperature stress	3.7.2	4.7.2
Vibration	3.7.3	4.7.3
Mechanical shock I	3.7.4.1	4.7.4.1
Mechanical shock II	3.7.4.2	4.7.4.2
Humidity	3.7.5	4.7.5
Subgroup 3		
Ripple voltage	3.6.7	4.6.7
Audio noise	3.6.8	4.6.8
Temperature operation at -17.8°C	3.7.1.2	4.7.1.2

4.5.3.3 Reinspection of conforming Group C samples. Unless otherwise specified, the samples which have been subjected to and passed Group C inspection shall be accepted on the contract provided all damages are repaired. Samples which were subjected to Humidity shall be baked at a minimum of 55°C and 400 mm Hg maximum for at least 24 hours prior to use.

4.5.4 Group D inspection. This inspection shall consist of the tests specified in Table IX and shall be performed on units from lots which have been subjected to and met Group A and Group B inspections. Tests conditions, methods, and procedures shall be in accordance with paragraph 4.9 and MIL-STD-781.

4.5.4.1 Sampling for Group D inspection. Sampling for Reliability shall be selected at random from each lot. Unless otherwise specified, for reliability testing purposes a lot is defined as one month's production. 10 power supplies from each of two lots, a total of 20 power supplies, shall be combined to form the reliability sample.

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4.5.4.2 Group D failure. Failure of the power supplies on test to meet the requirements of Table IX shall be considered a failure of the Group D inspection. Occurrance of pattern failure(s) (see 3.12.14) shall be considered a failure of the Group D inspection. Actions required in the event of a Group D failure shall be specified in the contract or purchase order (see 6.2.c).

TABLE IX. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.9	4.9

4.5.4.3 Reinspection of conforming Group D samples. Unless otherwise specified, the samples which have been subjected to and passed Group D inspection shall be accepted on the contract provided all damages are repaired.

4.6 Test methods.

4.6.1 Voltage adjustments.

4.6.1.1 Maximum gain.

a. Measure V_{10} with the adjustment R_1 and R_2 at their maximum gain setting with the conditions as specified in 3.6.1.1.a.

b. Measure V_{10} as the adjustment R_1 varies from its maximum to minimum gain setting while the adjustment R_2 remains at its maximum gain setting. The conditions of the power supply are specified in 3.6.1.1.b.

c. Measure V_{10} as the adjustment R_1 varies from its minimum to maximum gain setting while the adjustment R_2 remains at its maximum gain setting. The conditions of the power supply are specified in 3.6.1.1.c.

Failure to meet the requirements of 3.6.1.1 shall constitute failure of this test.

4.6.1.2 Minimum gain.

a. Observe and measure the voltage changes of V_{10} while the adjustment R_2 varies from its maximum to minimum gain setting at the conditions specified in 3.6.1.1.a. The setting of R_1 shall not be changed during the performance of this test.

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b. Observe and measure the voltage changes of V_{10} while the adjustment R_1 varies from its maximum to minimum gain setting at the conditions specified in 3.6.1.1.b. The setting of R_1 shall not be changed during the performance of this test.

Failure to meet the requirements of 3.6.1.2 shall constitute failure of this test.

4.6.2 ABC current characteristics.

4.6.2.1 ABC start current. Measure and record V_{10} with 3.0 Vdc input as the initial preset voltage. Calculate 90 percent, ±2.0 percent value points of the V_{10}.

a. Obtain the ABC start current value by increasing the ABC current source until the V_{10} falls to 90 percent, ± 2.0 percent of initial preset value with switch(es) closed as sepcified in Table X.

b. Repeat the step above with 2.0 Vdc input to the power supply.

Failure to meet the requirements of 3.6.2.1 shall constitute failure of this test.

4.6.2.2 ABC finish current. With the EGAC at the specified gain positions in Table X, the ABC finish current value shall be obtained varying the ABC source current until the V_{10} is reduced below 200 Vdc.

a. Obtain the ABC finish current value with 3.0 Vdc input and switch(es) closed as specified in Table X. Calculate the difference between ABC finish current value and ABC start current value of 4.6.2.1.a.

b. Obtain the ABC finish current value with 2.0 Vdc input and switch(es) closed as specified in Table X. Calculate the difference between ABC finish current value and ABC start current value of 4.6.2.1.b.

Failure to meet the requirements of paragraph 3.6.2.2 shall constitute failure of this test.

4.6.3 Output regulation.

4.6.3.1 Normal mode. Perform the measurements required for each of the output voltages as specified in Table XI. Failure to meet the requirements of 3.6.3.1 shall constitute failure of this test.

4.6.3.2 High light resolution mode. Perform the measurements required for each of the output voltages as specified in Table XII. Failure to meet the requirements of 3.6.3.2 shall constitute failure of this test.

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4.6.3.3 Bright source protection mode. Perform the measurements required for each of the output voltages as specified in Table XIII. Failure to meet the requirements of 3.6.3.3 shall constitute failure of this test.

4.6.4 Input current characteristics.

4.6.4.1 Maximum peak starting (MPS) and maximum peak steady state (MPSS). Perform the measurements for MPS and MPSS as specified in Table XIV as follows: All power supply outputs shall be completely discharged to ground before performing the test for MPS. Connect the current probe or equivalent to the dc input lead as in Figure 2. Apply the input voltage for the specified test condition and allow the storage scope or equivalent to display the peak current waveform. Sweep ground on the storage scope and read the maximum sweep point above ground as MPS. Read the peak to peak steady state signal as MPSS. The MPS displayed within the initial 100 milliseconds after the applied input voltage shall be excluded from the measurement. Failure to meet the requirements of 3.6.4.1 shall constitute failure of this test.

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TABLE X. ABC current characteristics test.

		Output	Switches	EGAC at max 3.0 Vdc Input	EGAC at max 2.0 Vdc Input	EGAC at min 3.0 Vdc Input	EGAC at min 2.0Vdc Input
Temperature	Measurement	Terminal	Closed				
+52°C	ABC Start current	13	S7	(*)	*	--	--
	ABC Finish current	13	S7	(*)	*	*2	*2
	ABC Start current	12	S6	*1	*1	--	--
	ABC Finish current	12	S6	*1	*1	*1	*1
+23°C	ABC Start current	13	S7	*	*	--	--
	ABC Finish current	13	S7	*	*	*2	*2
	ABC Start current	12	S6	*1	*1	--	--
	ABC Finish current	12	S6	*1	*1	*1	*1
-54°C	ABC Start current	13	S7	(*)	*	--	--
	ABC Finish current	13	S7	(*)	*	*2	*2
	ABC Start current	12	S6	*1	*1	--	--
	ABC Finish current	12	S6	*1	*1	*1	*1

(*) IPT test only
1 Group C Subgroup 1 inspection and IPT test only
*2 Group C Subgroup 2 inspection and IPT test only
* Test as required

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TABLE XI. Output regulation, normal mode tests.

Temp	Measurement	Switch(es) Closed	Load Current	EGAC at max 3.0 Vdc	EGAC at max 2.0 Vdc	EGAC at min 3.0 Vdc	EGAC at min 2.0 Vdc
+52°C	All output voltages	--	--	(*)	*	(*)	*
	V10	S6	35,± 5% of ABC start	(*)	*1	--	--
	V10	S7	35, ± 5% of ABC start	(*)	*2	--	--
+23°C	All ouput voltages	--	--	*	(*)	*	(*)
	V10	S6	35, ± 5% of ABC start	*1	--	--	--
	V10	S7	35, ± 5% of ABC start	*	(*)	--	--
-54°C	All output voltages	--	--	(*)	*	(*)	*
	V10	S6	35, ± 5% of ABC start	(*)	*1	(*)	--
	V10	S7	35, ± 5% of ABC start	(*)	*2	(*)	--

(*) IPT test only
*1 Group C Subgroup 1 inspection and IPT test only
*2 Group C Subgroup 2 inspection and IPT test only
* Test as required

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TABLE XII. Output regulation, high resolution mode tests.

Temp	Measurement	Switch(es) Closed	Load Current	EGAC at max 3.0 Vdc	EGAC at max 2.0 Vdc	EGAC at min 3.0 Vdc	EGAC at min 2.0 Vdc
+52°C	V5, V8, V10, V13	S3, S7	ABC Finish	(*)	*2	--	(*)
	V6, V7, V9 V10, V12	S4, S6	ABC Finish	(*)	*1	--	(*)
+23°C	V5, V8, V10 V13	S3, S7	ABC Finish	*	(*)	(*)	(*)
	V6, V7, V9 V10, V12	S4, S6	ABC Finish	*	(*)	(*)	(*)
-54°C	V5, V8, V10 V13	S3, S7	ABC Finish	(*)	*2	--	(*)
	V6, V7, V9 V10, V12	S4, S6	ABC Finish	(*)	*1	--	(*)

(*) IPT test only
*1 Group C Subgroup 1 inspection and IPT test only
*2 Group C Subgroup 2 inspection and IPT test only
* Test as required

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TABLE XIII. Output regulation, bright source protection mode tests.

Temp	Measurement	Switch(es) Closed	Load Current	EGAC at max 3.0 Vdc	EGAC at max 2.0 Vdc	EGAC at min 3.0 Vdc	EGAC at min 2.0 Vdc
+52°C	V5, V10, V13	S2	No load	(*)	*2	--	(*)
	V6, V10, V12	S5	No load	(*)	*1	--	(*)
+23°C	V5, V10, V13	S2	No load	*	(*)	(*)	(*)
	V6, V10, V12	S5	No load	*	(*)	(*)	(*)
-54°C	V5, V10, V13	S2	No load	(*)	*2	--	(*)
	V6, V10, V12	S5	No load	(*)	*1	--	(*)

(*) IPT test only
1 Group C Subgroup 1 Inspection and IPT test
*2 Group C Subgroup 2 Inspection and IPT test
* Test as required.

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TABLE XIV. Input current characteristics tests.

Temp	Measurement	Switch(es) Closed	Load Current	EGAC at max 3.0 Vdc	EGAC at max 2.0 Vdc	EGAC at min 3.0 Vdc	EGAC at min 2.0 Vdc
+52°C	MASS	--	--	(*)	*	--	--
	MASS	S6	35, ± 5% of ABC start current	(*)	*1	--	--
+23°C	MPS	--	--	(*)	--	--	--
	MPSS	--	--	(*)	--	--	--
	MASS	--	--	*	(*)	(*)	(*)
	MASS	S6	35, ± 5% of ABC start current	(*)	(*)	--	--
-54°C	MASS	--	--	(*)	*	--	--
	MASS	S6	35, ± 5% of ABC start current	(*)	*1	--	--

(*) IPT test only
*1 Group C Subgroup 1 inspection and IPT test only
* Test as required

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TABLE XV. Rise time test.

Temp	Photocathode current	Voltage output	EGAC at Max 3.0 Vdc	EGAC at Max 2.0 Vdc	EGAC at Min 3.0 Vdc	EGAC at Min 2.0 Vdc
+52°C	0.02 nA	V10	(*)	*	(*)	(*)
	0.12 - 0.24 uA	V10	(*)	(*)	(*)	(*)
+23°C	0.02 na	V10	(*)	(*)	(*)	(*)
	0.12 - 0.24 uA	V10	(*)	(*)	(*)	(*)
-17.8°C	0.02 nA	V10	*1	*1	--	--
	0.12 - 0.24 uA	V10	*1	*1	--	--
-54°C	0.02 nA	V10	(*)	*	(*)	(*)
	0.12 - 0.24 uA	V10	(*)	(*)	(*)	(*)

(*) IPT test only
*1 Group C Subgroup 3 inspection and IPT only; Input voltage to be provided by battery BA-5567/U or BA-3058/U or equivalent power source (see 3.12.15).
* Test as required

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4.6.4.2 Maximum average steady state (MASS). Perform the measurements for MASS as specified in Table XIV. Failure to meet the requirements of 3.6.4.2 shall constitute failure of this test.

4.6.5 Rise time. Connect the power supply to the intensifier module as shown in Figure 5. The rise time shall be verified by displaying V_{10} on a storage oscilloscope or equivalent. Sensitivity of the equipment shall detect a 20 Vdc overshoot. Perform the measurements of V_{10} as specified in Table XV. The tests shall be conducted as follows: Apply the illumination to the intensifier. Then apply the specified input potential to the power supply while monitoring V_{10}. Sweep the steady state voltage trace for V_{10}. Measure the time for V_{10} to settle within 10 Vdc of its steady state value. All outputs from the power supply shall be fully discharged from any previous operation prior to performing the rise time test. Failure to meet the requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Response time. Connect the power supply as shown in Figure 5. The response time shall be verified by displaying V_{10} on a storage oscilloscope or equivalent. Sensitivity of the equipment shall allow detection of a 20 Vdc overshoot. With the EGAC at maximum gain position apply the input voltage to the power supply and test the response time as stated in 4.6.6.1 and 4.6.6.2.

4.6.6.1 Recovery time, profile (a). Using a light source and input steady state ambient illumination that will produce photocathode current (see 3.12.13) not greater than 0.02 nA, apply the pulse input illumination to the intensifier module to produce photocathode current within the limits specified in Figure 3. Record the recovery time from the end of the applied pulse. Failure to meet the requirements of 3.6.6.1 shall constitute failure of this test.

4.6.6.2 Recovery time, profile (b). Using the conditions set forth in 4.6.6.1, apply the periodic pulsed input illumination to the intensifier module to produce photocathode current not less than the values specified in Figure 4, in such a manner that the third cycle ends on a simulated tracer pulse. Determine the recovery time from the end of each illumination pulse. Failure to meet the requirements of 3.6.6.2 shall constitute failure of this test.

4.6.7 Ripple voltage.

a. Disconnect all voltmeters from the power supply. With the EGAC at maximum gain position and R_4 at 75 megohms, apply an input voltage of 3.0 Vdc. Display the ripple voltage of each output using a Jennings Probe and storage oscilloscope or equivalent. Read the peak to peak trace as the ripple voltage. Calculate percent ripple voltage as follows:

$$\frac{\text{V ripple p-p}}{\text{Vdc output}} \times 100 \text{ percent}$$

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TABLE XVI. Operational stability test

Temp	Measurement	Photocathode current	EGAC at Max		EGAC at min	
			3.0 Vdc input	2.0 Vdc input	3.0 Vdc input	2.0 Vdc input
+52°C	V10, MASS input current	0.12 - 0.24 uA	(*)	*	(*)	(*)
+23°C	V10, MASS input current	0.12 - 0.24 uA	(*)	(*)	(*)	(*)
-17.8°C	V10, MASS input current	0.12 - 0.24 uA	*1	*1	*1	*1
-54°C	V10, MASS input current	0.12 - 0.24 uA	(*)	*	(*)	(*)

(*) IPT test only.
*1 Group C Subgroup 3 inspection and IPT inspection only;
Input voltage to be provided by battery BA-5567/U or BA-3058/U or an equivalent power source (see 3.12.15)
* Test as required

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b. To test the ripple voltage during the bright source protection mode, connect a 1 megohm resistor between terminal 5 and 11 and display the ripple voltage for V_s using Jennings Probe and storage oscilloscope or equivalent. Calculate the peak to peak ripple voltage for V_s Disconnect the 1 megohm resistor between the terminal 5 to 11 and connect it between terminals 6 and 7. Simultaneously display the ripple voltages from terminal 6 to 11 and 7 to 11 and calculate the difference on the storage oscilloscope or equivalent.

Failure to meet the requirements of 3.6.7 shall constitute failure of this test.

4.6.8 Audio noise. The power supply shall be set up for operation within the image intensifier housing in an area where the background noise level is at least 10 dB below the sound level to be measured. The power supply shall then be placed in operation and sound pressure measurements made from a distance of 6 inches on each side and at the same height as the power supply centerline; at the ends of each monocular along the equipment centerline; and above and below the power supply at the vertical centerline. Measurement shall be made with the equipment operating, using an octave analyzer whose characteristics comply with ANSI Specification S1.2-1962. There shall be no observer, instrument or reflecting surface, other than the floor within one meter of the test item during the measurements. The sound pressure levels measures at 6 inches (dB reference 0.0002 microbars) shall be recorded for the frequencies listed below.

Frequency		
Center Frequency (Hz)	Octave Band (Hz)	Maximum Sound Pressure Level (dB: ref 0.0002 microbars
8000	5600-11200	17
16000	11200-22400	16

Verification of the operating frequencies shall be determined during the ripple voltage test (see 4.6.7).

Failure to meet the requirements of 3.6.8 shall constitute failure of this test.

4.6.9 Operational stability. Connect the power supply to the intensifier module as shown in Figure 5. The operational stability shall be verified by (a) observing for arcing and monitoring MASS input current, and (b) displaying V_{10} on a storage oscilloscope or equivalent for a period of no less than 30 seconds. Sensitivity of the equipment shall allow detection of a 0.5 mA and 5 volt fluctuations. Perform the measurements specified in Table XVI for input current and V_{10} as follows: Apply the specified input voltage to the power supply. Apply the specified step change

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of input illumination to the photocathode of the intensifier module. Within one (1) second after application of the input illumination, monitor V_{10} and MASS current for not less than 30 seconds. The rise time of the input illumination step changes shall be less than 250 milliseconds. Failure to meet the requirements of 3.6.9 shall constitute failure of this test.

4.6.10 Operational safety.

4.6.10.1 Short circuit protection. Operate the power supply. While operating, directly short all outputs to ground for a least one minute. Remove the short and inspect the power supply for damage. Failure to meet the requirement of 3.6.10.1 shall constitute failure of this test.

4.6.10.2 Reverse polarity. Apply a reverse polarity of 3.0 volts dc input to the power supply for at least 10 minutes. Failure to meet the requirements of 3.6.10.2 shall constitute failure of this test.

4.6.11 Electromagnetic interference. The power supply shall be tested under equipment classification Class A3 with test requirements RE01, RE02, RS01, RS02 and RS03 as stipulated in MIL-STD-461. The tests shall be performed in accordance with MIL-STD-462 except that the distance for measurement shall be 42 cm instead of 7 cm for test method RE01. The power supply shall be tested on a simulated intensifier module, and inside the tube housing (per drawing Housing, Image Intensifier, SM-D-850299) operated at ground potential. MASS input current shall be measured during the radiation susceptibility tests. Failure to meet the requirements of 3.6.11 shall constitute failure of this test. The frequency to be tested under RE02 shall be 14KHz - 10GHz. The limitations on RE02 are as follows:

20.5 - 35.5KHz - 35dB
35.5 - 88.0KHz - 45dB
88.0 - 130.0KHz - 25dB
130.0 - 340KHz - 20dB

The Radiated Susceptibility frequencies under RS03 are as follows:

a. 10KHz - 2MHz
b. 2MHz - 30MHz
c. 30MHz - 10GHz
d. 10GHz - 40GHz

4.6.12 Weight. The requirement of 3.2 shall be met using a scale or balance calibrated to ± .1 gram.

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4.7 Environmental conditions.

4.7.1 Temperature operation.

4.7.1.1 Temperature operation at +52°C. Place the power supply in a test chamber at room temperature (see 3.12.12). Raise the temperature of the tests chamber to +52°C and hold at this temperature for a minimum of one hour. At the end of the one hour stabilizing period, the following tests shall be performed:

a. ABC current as specified in Table X. Test procedure for ABC current shall be in accordance with 4.6.2.

b. Output regulation as specified in Table XI, XII, and XIII. The test procedure for output regulation shall be in accordance with 4.6.3.

c. MASS input current as specified in Table XIV. The test procedure for input current shall be in accordance with 4.6.4.2.

d. Rise time as specified in Table XV. The test procedure for rise time shall be in accordance with 4.6.5.

e. Operational stability as specified in Table XVI. The test procedure for operational stability shall be in accordance with 4.6.9.

Failure to meet the requirements of 3.7.1.1 shall constitute failure of this test.

4.7.1.2 Temperature operation at -17.8°C. After the +52°C tests are completed, remove the input voltage to the power supply. Lower the temperature of the test chamber to room temperature (see 3.12.12) and connect a BA-3058/U or BA-5567/U to the power supply and place in the test chamber. Lower the temperature of the test chamber to -17.8°C. Hold this temperature for a minimum of one hour. At the end of this one hour period, measure the rise time and operational stability specified in Table XV and XVI. Repeat the preceding test using an equivalent power source (see 3.12.15). Failure to meet the requirements of 3.7.1.2 shall constitute failure of this test.

4.7.1.3 Temperature operation at -54°C. After testing at +52°C (and -17.8°C for IPT), remove the input voltage to power supply and lower the temperature of test chamber to -54°C and hold at this temperature for a minimum of one hour. At the end of the one hour stabilizing period, perform the tests for output regulation, ABC current characteristics, rise time, MASS input current, and operational stability as sepcified in Table X through XVI using the same test procedures as 4.7.1.1. At the conclusion of the tests, remove the input voltage to the power supply. Failure to meet the requirements of 3.7.1.3 shall constitute failure of this test.

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4.7.2 Environmental tempeature stress.

4.7.2.1 High temperature stress. Test the power supply in accordance with MIL-STD-810, method 501.1. The following details shall apply:

a. Procedure Number: Procedure I

b. Pretest performance record: Output regulation (4.6.3), ABC current characteristics (4.6.2), MASS input current (4.6.4.2).

c. Performance record at temperature - Step 5: Temperature operation at +52°C (4.7.1.1).

d. Step 2: Storage temperature at +68°C.

e. Step 3: Storage time - not less than 2 hours.

f. Step 4: Operating temperature +52°C.

g. Steps 7 and 8 to be omitted.

Failure to meet the requirements of 3.7.2.1 shall constitute failure of this test.

4.7.2.2 Low temperature stress. Test the power supply in accordance with MIL-STD-810, Method 502.1. The following details and exceptions shall apply:

a. Pretest Performance Record: Output regulation (4.6.3), ABC current characteristics (4.6.2), MASS input current (4.6.4.2).

b. Performance record at temperature - Step 5: Temperature operation at -54°C (4.7.1.3).

c. Storage temperature, Step 2: -62°C for not less than 2 hours.

d. Step 3: not applicable.

e. Lowest operating temperature, Step 4: -54°C.

f. Steps 7 and 8 to be omitted.

Failure to meet the requirements of 3.7.2.2 shall constitute failure of this test.

4.7.2.3 Temperature shock. Test the power supply in accordance with MIL-STD-810, Method 503.1. The following details and exceptions shall apply:

a. Pretest Performance Record: Output regulation (4.6.3), ABC current characteristics (4.6.2), MASS input current (4.6.4.2).

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b. Temperature extremes: +68°C and -62°C.

Failure to meet the requirements of 3.7.2.3 shall constitute failure of this test.

4.7.3 Vibration. The input voltage shall not be applied during vibration. Tolerances on specified frequencies shall be ± 1 percent. Mount the power supplies rigidly. Subject the power supplies to simple harmonic motion applied in planes parallel and perpendicular to the radial axis. In 30 minutes vary the frequency from 10 Hz to 3500 Hz and return to 10 Hz in a logarithmic sweep. Perform the frequency sweep twice in each plane. Failure to meet the requirements of 3.7.3 shall constitute failure of this test.

4.7.4 Mechanical shock.

4.7.4.1 Mechanical shock I. The input voltage shall not be applied during the shock. The power supply shall be rigidly mounted in approved test fixture. Subject the power supply to the shock impulse specified in 3.7.4. Test shall be monitored with an accelerometer mounted on the test fixture near the power supply. Pulse shape and amplitudes shall be monitored. Failure to meet the requirements of 3.7.4 shall constitute failure of this test.

4.7.4.2 Mechanical shock II. This test is to be conducted in the same manner as described in 4.7.4.1. Subject the power supply to the shock impulse specified in 3.7.4.2. Failure to meet the requirements of 3.7.4.2 shall constitute failure of this test.

4.7.5 Humidity. Prior to starting the test, the internal chamber temperature shall be at ambient condition with uncontrolled humidity, Place the power supply in the test chamber and increase the chamber condition to a relative humidity of not less than 95 percent at +52°C. Maintain at this condition for a period of not less than 4 hours. Remove the test item from the chamber and perform the following mesurements: ABC current characteristics (4.6.2), output regulation (4.6.3), MASS input current (4.6.4.2), rise time (4.6.5) and operational stability (4.6.9). Prior to measurements, excess moisture may be removed from the exterior surfaces by wiping the external surface only. Failure to meet the requirements of 3.7.5 shall constitute failure of this test.

4.8 Burn-in/ESS. Subject the power supply to 24 continuous cycles of operations at +52°C. Each cyclic operation shall consist of 55 minutes ON-time and 5 minutes OFF-time. All outputs shall be at no load during the operation. Verification of this test shall be accomplished by audit of the burn-in record.

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4.9 Reliability.

4.9.1 Reliability testing. The initial production and quality conformance reliability inspections shall be conducted in accordance with MIL-STD-781 and Test Plan VI-D of MIL-HDBK-781. The detailed test conditions, procedures, and methods shall be in accordance with the approved Reliability Test Procedure.

4.9.2 Reliability test procedure. The power supply shall be set at the high voltage operating setting prior to subjection to a reliability test cycle as specified in 4.9.2.1. Sufficient reliability test cycles shall be performed to satisfy the requirements of the applicable Test Plan. The number of operating hours for all units shall be continuously monitored and recorded. Failure of a power supply to meet the requirements of Table XVII shall be considered a failure and the hours accumulated by the power supplies to the last successful check shall be used in the totalized hours for the lot under test. At the conclusion of the test, each power supply shall be submitted to and pass the inspection of Table XVII. In addition, during IPT reliability testing, each power supply shall be submitted to the inspection of Table XVII at least once every 400 hours.

4.9.2.1 Reliability test cycle. A reliability test cycle shall consist of 24 duty cycles of 55 minutes ON-time and 5 minutes OFF-time at +52°C, followed by a non-operating temperature (4.9.2.1.1), and then followed by 12 repetitive stress cycles of high light stress (4.9.2.1.2) and of bright source stress (4.9.2.1.3) sequentially at +52°C. The input voltage shall be 3.0 Vdc with EGAC at maximum gain position. The load at output 10 shall be 56 megohms.

4.9.2.1.1 Non-operation temperature. The non-operating temperature shall consist of changing the power supply temperature from +52°C to -17.8°C. The transition time between temperatures shall be within an half hour. The storage time at -17.8°C shall be minimum of a half hour.

4.9.2.1.2 High light stress. A cycle of this stress shall consist of the following load condition sequence:

a. 5 seconds at load condition 1.

b. 10 seconds at load condition 2.

c. 5 seconds at load condition 3.

d. 10 seconds at load condition 2.

The loads for each of these conditions are as follows:

Load condition 1-7 gigohm resistor between outputs 12 and 11, 200 megohm resistor between outputs 5 and 10, 56 megohm resistor between outputs 10 and 11, and no load at all other outputs.

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Load condition 2-56 megohm resistor between outputs 10 and 11, and no load for all other outputs.

Load condition 3-7 gigohm resistor between outputs 12 and 11, 160 megohm resistor between outputs 6 and 10, 56 megohm resistor between outputs 10 and 11, and no load for all other outputs.

4.9.2.1.3 Bright source stress. A cycle of this stress shall consist of the following load condition sequence:

a. 3 seconds at load condition 1.

b 10 seconds at load condition 2.

c. 3 seconds at load condition 3.

d. 10 seconds at load condition 2.

The loads for each of these conditions are as follows:

Load condition 1- 1 megohm resistor between outputs 5 and 11, 56 megohm resistor between outputs 10 and 11, and no load for all other outputs.

Load condition 2- 56 megohm resistor between outputs 10 and 11 and no load for all other outputs.

Load condition 3- 1 megohm resistor between outputs 6 and 7, 56 megohm resistor between outputs 10 and 11, and no load for all other outputs.

TABLE XVII. Reliability performance tests.

Inspection	Requirement paragraph	Inspection paragraph
ABC current characteristics	3.6.2	4.6.2
Output regulation	3.6.3	4.6.3
Operational stability	3.6.9	4.6.9
MASS input current	3.6.4.2	4.6.4.2

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2.g).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1. Intended use. The power supply is intended to be used in the Image Intensifier, Night Vision, MX-9644/UV. The power supply for the Image Intensifier, Night Vision MX-9644/UV in accordance with MIL-I-49040 herein after referred to as the power supply, shall be a solid state electronic device to convert an unregulated 2.0 to 3.0 volts direct current (Vdc) input to multiple high voltage dc outputs.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification and amendments thereto.

b. Quantity and schedule for IPT testing for all line items.

c. Necessary actions by the contractor in the event of a Group C failure (see 4.5.3.2) and/or a Group D failure (see 4.5.4.2).

d. Sampling plans for Group B (4.5.2) and Group C (4.5.3.1) inspections and tests. For guidance unless otherwise specified, the sampling shall be in accordance with MIL-STD-105.

e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2, and 2.2).

f. MIL-STD-810C is used for environmental tests which specify MIL-STD-810.

g. Packing/Packaging requirements.

6.3 Qualification. With respect to products requiring qualification, unless otherwise specified by the contract or purchase order, if a product is not qualified at time of award, the contract shall require qualification prior to first delivery. The activity responsible for approving qualified products is the CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703. Information pertaining to qualification of products may be obtained from the above activity.

6.4 Definitions. See 3.12.

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6.5 Subject term keyword listing.

Image intensifier
Microchannel wafer
Power supply

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CUSTODIAN:
Army - CR

USER ACTIVITY:
Army - MU

PREPARING ACTIVITY:
Army - CR

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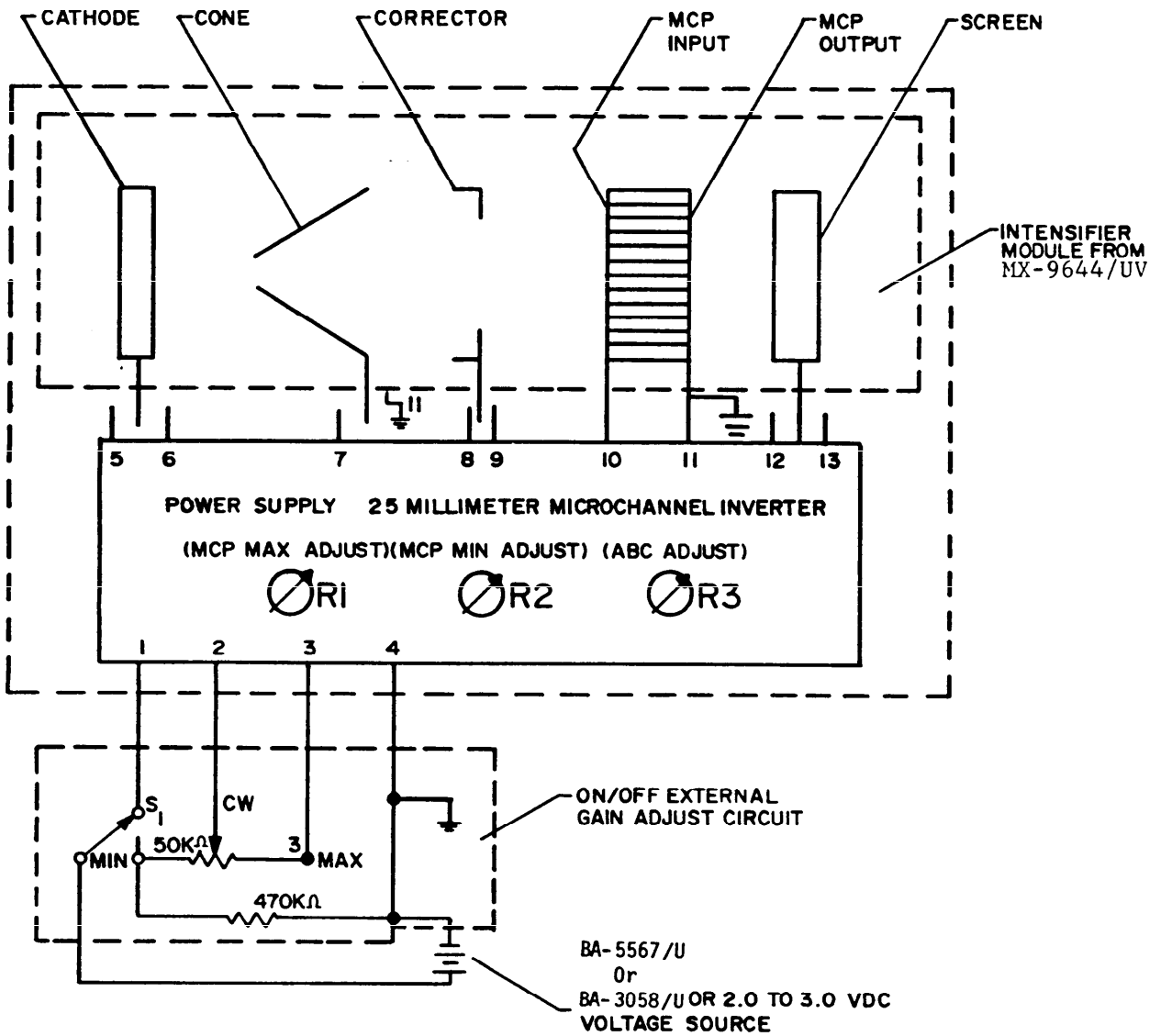


FIGURE 1 ELECTRICAL INTERFACE WIRING PROCEDURE

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POWER SUPPLY INPUT CONNECTIONS		
INPUT	LEAD COLOR CODE	CONNECTION DESIGNATION
1	RED	DC INPUT, POSITIVE B
2	BLUE	EXTERNAL GAIN ADJUST WIPER
3	ORANGE	EXTERNAL GAIN ADJUST, MAX SIDE
4	BLACK	GROUND

POWER SUPPLY OUTPUT CONNECTIONS		
OUTPUT	CONNECTION DESIGNATION	NOTE REF
5	CATHODE	NOTE 1
6		
7	CONE	NOTE 1
11		
8	CORRECTOR	NOTE 1
9		
10	MCP INPUT	
11	MCP OUTPUT	
12	SCREEN	
13		

ELECTRICAL INTERFACE WIRING PROCEDURE NOTES:

1. SELECT OUTPUTS TO ACHIEVE RESOLUTION REQUIREMENT PER MIL-I-49040. SELECTION OF OUTPUT 9 SHALL DICTATE THAT OUTPUTS 6, 7, AND 12 BE USED. LIKEWISE, SELECTION OF OUTPUT 8 DICTATES OUTPUTS 5, 11, AND 13 BE USED.

FIGURE 1

ELECTRICAL INTERFACE WIRING PROCEDURE CONTINUED

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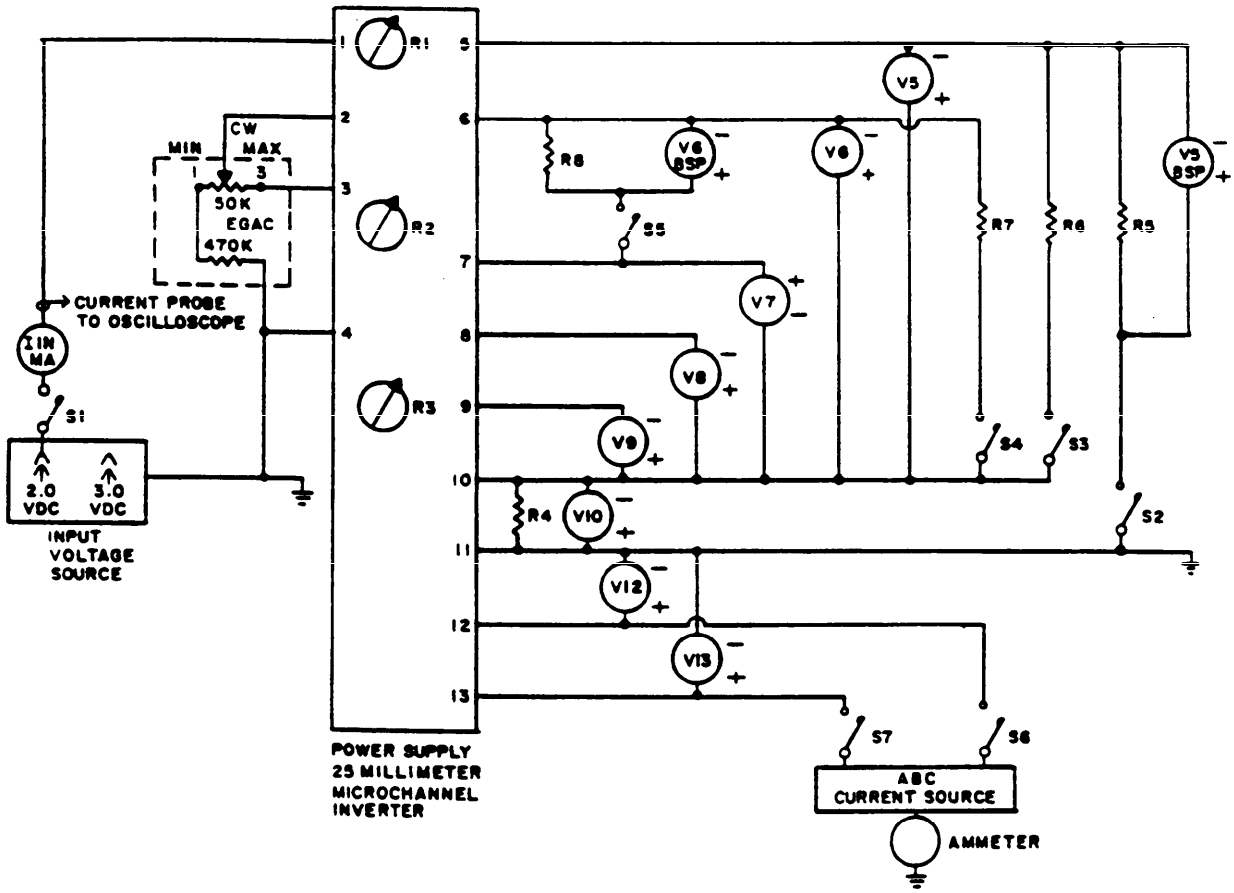


FIGURE 2 POWER SUPPLY TEST CIRCUIT

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NOTES:

A. RESISTANCE R_4 SHALL VARY ITS VALUE AS SPECIFIED BELOW:

RESISTANCE R_4

POWER SUPPLY SETTING	AMBIENT	-54°C	+52°C
HIGH VOLTAGE	75 MEGOHMS	160 MEGOHMS	56 MEGOHMS
STANDARD	75 MEGOHMS	160 MEGOHMS	56 MEGOHMS
LOW VOLTAGE	350 MEGOHMS	750 MEGOHMS	260 MEGOHMS

THE RESISTANCE VALUES SPECIFIED FOR R_4 SHALL INCLUDE THE EFFECT OF THE IMPEDANCE OF VOLTMETER, V_{10}. TOLERANCE ON SPECIFIED VALUES FOR R_4 SHALL BE ± 2 PERCENT.

B. VOLTMETERS; V_5, V_6, V_7, V_8, V_9, V_{12}, AND V_{13} FOR MEASURING OUTPUTS 5, 6, 7, 8, 9, 12, AND 13 RESPECTIVELY, SHALL BE ELECTROSTATIC VOLTMETERS OR EQUIVALENT. TOTAL LOADING EFFECT OF THESE VOLTMETERS AND TEST CIRCUIT WITH ALL SWITCHES IN THE OPEN POSITION SHALL BE LESS THAN 25 NANOAMPERES AT EACH OUTPUT.

C. RESISTORS SHALL BE SELECTED SUCH THAT THE CURRENT FLOW AT EACH SWITCH IS AS FOLLOWS:

(1) R_6 WITH S_3 CLOSED, THE CURRENT AT S_3 SHALL NOT BE LESS THAN 2.5 MICROAMPERES.

(2) R_7 WITH S_4 CLOSED, THE CURRENT AT S_4 SHALL NOT BE LESS THAN 2.5 MICROAMPERES.

D. RESISTORS R_5 AND R_8 SHALL BE 1.0 MEGOHMS, ± 2 PERCENT, AND SHALL INCLUDE THE EFFECT OF THE IMPEDANCE OF RESPECTIVE VOLTMETER.

FIGURE 2
POWER SUPPLY TEST CIRCUIT --
CONTINUED

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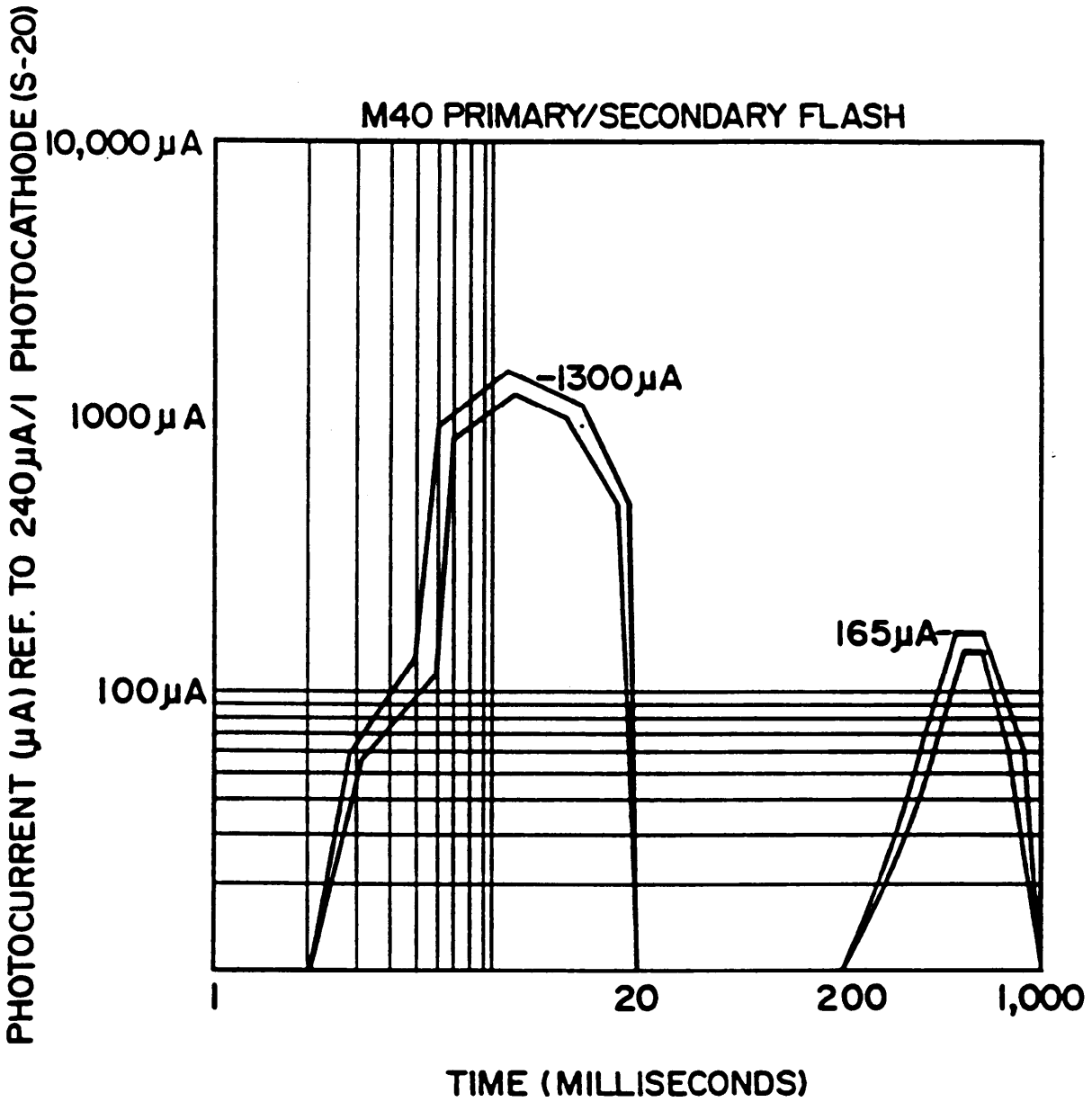


FIGURE 3 PROFILE(a)

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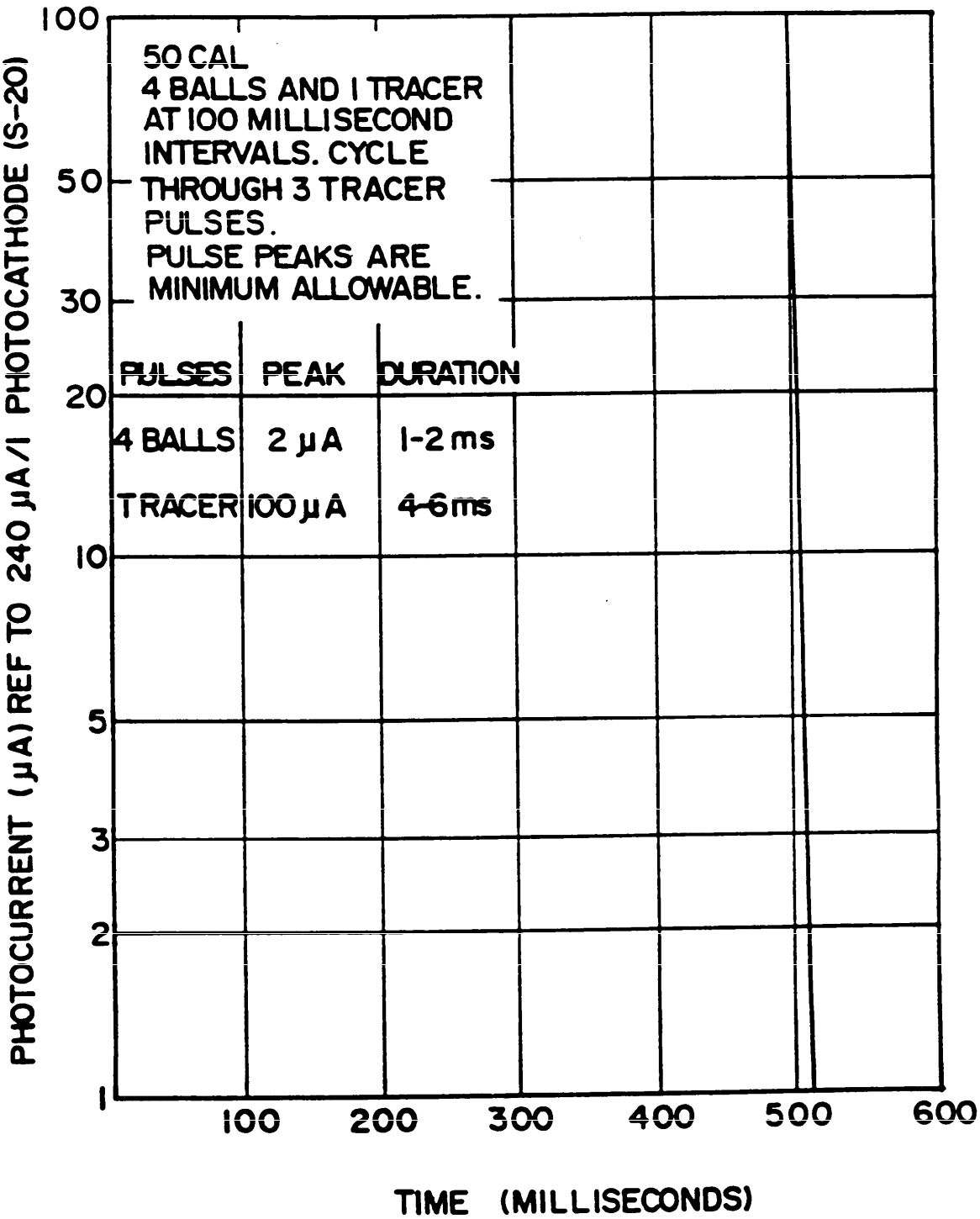


FIGURE 4 PROFILE (b)

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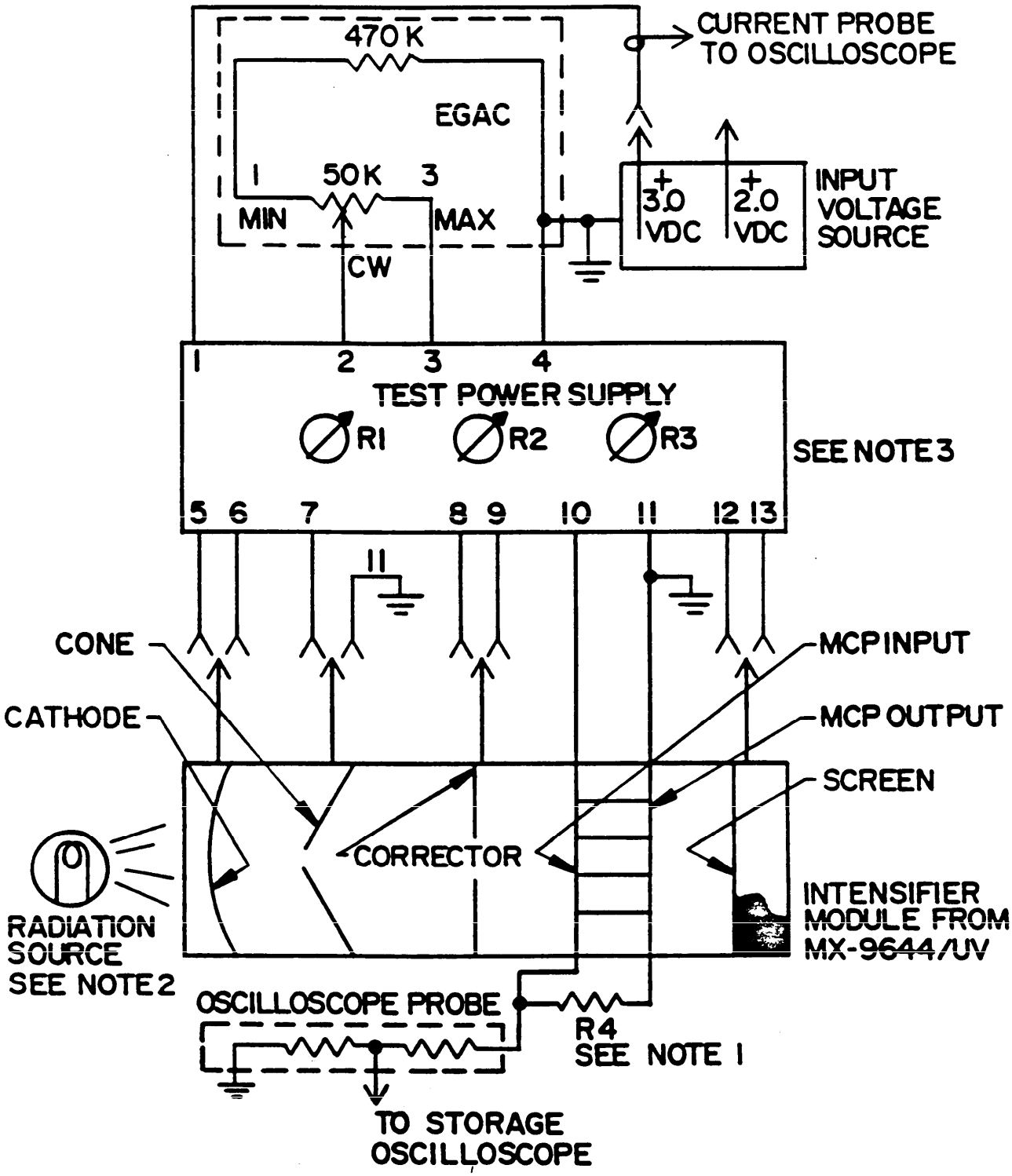


FIGURE 5
RISE-TIME, RESPONSE-TIME, AND OPERATIONAL STABILITY
TEST CIRCUIT

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NOTES:

1. THE RESISTANCE VALUES FOR R_4 SHALL INCLUDE THE EFFECT OF THE IMPEDANCE OF THE INTENSIFIER AND THE SCOPE PROBE. R_4 SHALL BE NOT GREATER THAN THE VALUES AS SPECIFIED BELOW:

RESISTANCE R_4

POWER SUPPLY SETTING	AMBIENT	-54°C	+52°C
HIGH VOLTAGE	75 MEGOHMS	160 MEGOHMS	56 MEGOHMS
STANDARD	75 MEGOHMS	160 MEGOHMS	56 MEGOHMS
LOW VOLTAGE	350 MEGOHMS	750 MEGOHMS	260 MEGOHMS

2. THE RADIATION SOURCE USED IN THE TEST SHALL PRODUCE PHOTOCATHODE CURRENT IN ACCORDANCE WITH THE TEST REQUIREMENTS OF THE PARAGRAPHS 4.6.5, 4.6.6, AND 4.6.9.

3. CONNECT OUTPUTS TO INTENSIFIER TO ACHIEVE RESOLUTION REQUIREMENT PER MIL-I-49040 (EL). SELECTION OF OUTPUT 9 SHALL DICTATE THAT OUTPUTS 6, 7, AND 12 BE USED. LIKEWISE, SELECTION OF OUTPUT 8 DICTATES THAT OUTPUTS 5, 11, AND 13.

FIGURE 5 RISE-TIME, RESPONSE-TIME, AND OPERATIONAL STABILITY-TEST CIRCUIT -- CONTINUED

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-49090D(CR)	26 May 1992

3. DOCUMENT TITLE
POWER SUPPLY 25 MILLIMETER, MICROCHANNEL INVERTER

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
CECOM	(908) 532-5851	992-5851
c. ADDRESS (Include Zip Code) AMSEL-ED-TM FOrt Monmouth, NJ 07703-5023	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340. AUTOVON 289-2340	

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